

Tyrosine kinase inhibitor (TKI) maintenance therapy versus no TKI maintenance in adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation: a systematic review

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Abstract

Background: Multiple studies have examined tyrosine kinase inhibitor (TKI) maintenance therapy in adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation, yet a synthesised view of the evidence is needed to inform clinical decision-making. The relationship between the intervention and patient-relevant outcomes (['relapse-free survival', 'overall survival', 'risk of graft-versus-host disease']) remains incompletely characterised.

Methods: This systematic review followed a PRISMA-aligned protocol. PubMed was searched using a retrieval-augmented MeSH expansion strategy: an LLM extracted narrow primary terms from the PICO question, fetched seven reference papers, and emitted a three-axis Boolean query (conditions AND treatments AND outcomes). The top 100 records by PubMed Best Match relevance were screened by an LLM-driven agent against PICO-derived eligibility criteria. Studies meeting the relevance threshold underwent automated study-characteristics and outcome extraction, and were synthesised qualitatively.

Results: Of 100 records retrieved, **9 studies** met the eligibility criteria and contributed to the synthesis. Nine studies directly addressing the planned comparison found that tyrosine kinase inhibitor maintenance therapy generally improves relapse-free survival and overall survival compared to no TKI maintenance in adults with FLT3-ITD-mutated acute myeloid leukemia who have undergone allogeneic stem-cell transplantation, with no clear impact on graft-versus-host disease.

Conclusions: This automated synthesis suggests directions for further investigation. Findings should be triangulated with expert-curated systematic reviews and confirmed in primary studies where possible.

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Keywords: adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation, tyrosine kinase inhibitor (TKI) maintenance therapy, no TKI maintenance

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Background

adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation presents a substantial clinical and public-health burden. Existing therapeutic strategies vary in their efficacy and safety profiles, and high-quality syntheses of the available evidence are critical for clinical decision-making, guideline development, and the prioritisation of further research.

The role of tyrosine kinase inhibitor (TKI) maintenance therapy in this population has been the focus of multiple primary studies, but findings have not always been concordant. A comprehensive synthesis is therefore required to characterise the consistency, magnitude, and applicability of the evidence.

The clinical question motivating this review is: “In adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation, does tyrosine kinase inhibitor (TKI) maintenance therapy improve relapse-free survival and overall survival compared to no TKI maintenance, and what is its impact on the risk of graft-versus-host disease?”. The PICO framework (Population, Intervention, Comparator, Outcomes) was used to structure both the search strategy and the eligibility criteria, as detailed in the Methods.

Methods

This systematic review was conducted under the guidance of the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement. The retrieval, screening, and synthesis stages were executed by the REASON automated pipeline; methods detail follows.

Literature search strategy

Search terms were generated using a retrieval-augmented MeSH expansion strategy. An initial language-model call extracted three to four narrow primary terms from the PICO question; these terms were used to fetch seven on-topic reference papers from PubMed via the NCBI E-utilities API. A second language-model call, grounded in the titles and abstracts of those reference papers, generated an expanded term list across three axes (conditions, treatments, outcomes). The final query is constructed as a fat-OR within each axis joined by Boolean AND across axes, with no explicit field qualifier so that PubMed's automatic term mapping engages. The exact query string appears in the Appendix.

PubMed/MEDLINE was queried via the NCBI E-utilities `research` endpoint with `sort=relevance` and `retmax=100`. No explicit date or language restrictions were applied at the search step.

Inclusion and exclusion criteria

Inclusion and exclusion criteria were generated automatically from the PICO framework and applied during the screening stage. The criteria below were used:

Data collection and quality assessment

Study-level characteristics (study name, design, year, geographical setting, population, intervention, comparator, sample size, age) and per-arm outcome data were extracted automatically using language-model extraction agents. Extraction was performed independently from screening to limit confirmation bias on the part of the agent. Per-study extracted records are preserved as CSV artefacts and reproduced in Tables 1 and 2.

Quality of evidence was not formally graded by the automated pipeline; the reader is referred to standard tools (e.g., the Newcastle-Ottawa scale or Cochrane Risk of Bias 2) for follow-on assessment.

Data analyses

Records with cumulative screening score ≥ 3 were considered eligible. Given likely heterogeneity in study designs, populations, and outcome definitions, a narrative synthesis was performed; no formal meta-analysis was conducted. The synthesis agent integrated study-level characteristics and extracted outcomes into a structured discussion.

Results

Literature search results

The literature search retrieved 100 records. After automated screening against the predefined eligibility criteria, 91 records were excluded and **9 studies were included in the synthesis**. The selection process is summarised in Figure 1 (PRISMA flow).

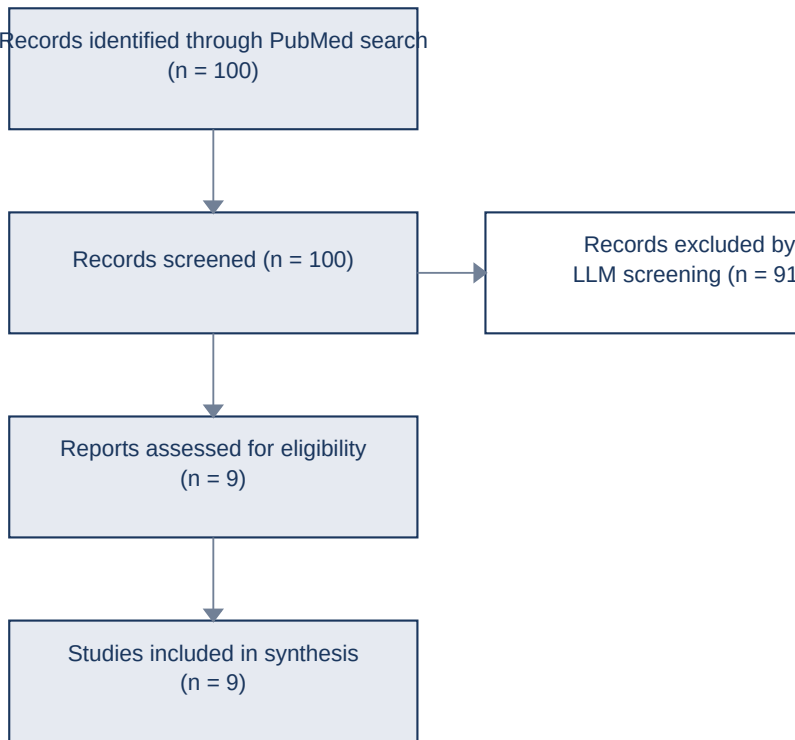


Fig. 1 Flowchart of study-selection process (PRISMA-2020).

Characteristics of the identified studies

Information was retrieved from the 9 studies included in this synthesis. The main characteristics of the eligible studies are summarised in Table 1.

Table 1 Main characteristics of the eligible studies

PMID	Author	Year	Setting	Population	Intervention	Comparator	n
38471061				Adults with FLT3-ITD AML in first remission who underwent allogeneic hematopoietic cell transplantation (HCT).	Gilteritinib maintenance (120 mg once daily for 24 months)	Placebo	356
39775763				Adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone hematopoietic cell transplantation (HCT).	gilteritinib maintenance therapy	placebo	
32673171				Adult patients with FLT3-ITD-positive acute myeloid leukemia in complete hematologic remission after allogeneic hematopoietic stem cell transplantation.	Sorafenib maintenance therapy	Placebo	83
40079105				Newly diagnosed FLT3-ITD-positive acute myeloid leukemia patients who underwent allogeneic hematopoietic cell transplantation in first complete remission (CR1) or composite CR1 (CRc1).	Quizartinib maintenance	Placebo maintenance	196
40590852				Patients with FLT3-ITD-mutated acute myeloid leukemia (AML) undergoing allogeneic hematopoietic cell transplantation (HCT).	Gilteritinib maintenance therapy	Placebo maintenance	
37414062			China	Adults aged 18-60 years with FLT3-ITD acute myeloid leukaemia who underwent allogeneic HSCT, had ECOG PS 0-2, composite complete remission before and after transplantation, and hematopoietic recovery within 60 days post-transplant.	Sorafenib maintenance	No TKI maintenance	202
37704613				Patients with FLT3-ITD acute myeloid leukemia (AML) undergoing allogeneic hematopoietic stem cell transplantation.	sorafenib maintenance	no sorafenib maintenance	613
40577201				FLT3-mutated acute myeloid leukemia patients who underwent allogeneic hematopoietic stem cell transplantation.	Sorafenib maintenance	No TKI maintenance	48
32791048			China	Adults (18-60 years) with FLT3-ITD acute myeloid leukemia who underwent allogeneic hematopoietic stem-cell transplantation, had composite complete remission before and after transplantation, and hematopoietic recovery within 60 days post-transplanta...	Sorafenib maintenance (400 mg orally twice daily)	Non-maintenance (control)	202

Outcomes extracted from included studies

Outcome measures extracted from each included study are presented in Table 2. The pipeline's outcomes CSV stores a JSON array of outcomes per study; the table below explodes that array so each row represents one (study × outcome) pair.

Table 2 Per-study outcome measures extracted by the pipeline

PMID	Comparison	Outcome	Effect	Value	95% CI	Follow-up
38471061	gilteritinib vs placebo	relapse-free survival	HR 0.679 (95% CI 0.459 to 1.005)	0.679	0.459–1.005	not reported
38471061	gilteritinib vs placebo	overall survival	not reported			not reported
38471061	gilteritinib vs placebo	risk of graft-versus-host disease	not reported			not reported
39775763	gilteritinib maintenance vs placebo	relapse-free survival	Generally, gilteritinib maintenance therapy was associated with improved relapse-free survival (RFS) for participants with detectable peri-HCT MRD, whereas no benefit was evident for those lacking detectable MRD. No overall effect estimate (HR, RR,...			not reported
39775763	gilteritinib maintenance vs placebo	overall survival	Mentioned that 14.8% of participants with multiple FLT3-ITD clones had worse survival irrespective of treatment arm, but no specific effect estimate (HR, RR, OR) reported for gilteritinib vs placebo.			not reported
39775763	gilteritinib maintenance vs placebo	risk of graft-versus-host disease	not reported			not reported
32673171	sorafenib vs placebo	Relapse-free survival (primary endpoint)	HR 0.39 (95% CI, 0.18 to 0.85)	0.39	0.18–0.85	median follow-up 41.8 months
32673171	sorafenib vs placebo	24-month Relapse-free survival probability	HR 0.256 (95% CI, 0.10 to 0.65)	0.256	0.1–0.65	24 months
32673171	sorafenib vs placebo	overall survival	not reported			not reported
32673171	sorafenib vs placebo	risk of graft-versus-host disease	not reported			not reported
40079105	quizartinib vs placebo	Overall Survival (quizartinib treatment in CR1)	HR 0.553 (95% CI 0.383-0.798)	0.553	0.383–0.798	not reported
40079105	quizartinib vs placebo	Overall Survival (quizartinib treatment in CRc1)	HR 0.645 (95% CI 0.470-0.886)	0.645	0.47–0.886	not reported
40079105	quizartinib vs placebo	relapse-free survival	not reported			not reported
40079105	quizartinib vs placebo	risk of graft-versus-host disease	Patients who underwent allo-HCT experienced post-transplant-related complications, mostly grade ≥2 graft-versus-host disease, as expected. No comparative effect estimate between quizartinib and placebo arms reported.			not reported

PMID	Comparison	Outcome	Effect	Value	95% CI	Follow-up
40590852	gilteritinib vs placebo	relapse-free survival	NPM1 comutation was associated with the largest magnitude of relapse-free survival benefit from post-HCT gilteritinib, but no specific effect estimate for gilteritinib vs placebo is reported.			not reported
40590852	gilteritinib vs placebo	overall survival	not reported			not reported
40590852	gilteritinib vs placebo	risk of graft-versus-host disease	not reported			not reported
37414062	sorafenib maintenance vs non-maintenance	5-year overall survival	HR 0.55 (95% CI 0.34-0.88)	0.55	0.34–0.88	5-year follow-up (median 60.4 months)
37414062	sorafenib maintenance vs non-maintenance	5-year leukaemia-free survival	HR 0.47 (95% CI 0.30-0.73)	0.47	0.3–0.73	5-year follow-up (median 60.4 months)
37414062	sorafenib maintenance vs non-maintenance	5-year cumulative incidence of relapse	HR 0.33 (95% CI 0.18-0.60)	0.33	0.18–0.6	5-year follow-up (median 60.4 months)
37414062	sorafenib maintenance vs non-maintenance	5-year cumulative incidence of chronic GVHD	HR 0.82 (95% CI 0.56-1.19)	0.82	0.56–1.19	5-year follow-up (median 60.4 months)
37704613	sorafenib maintenance vs no sorafenib maintenance	3-year overall survival	HR 0.50 (95% CI 0.37-0.69)	0.5	0.37–0.69	3-year post-transplantation; median follow-up 36.5 months
37704613	sorafenib maintenance vs no sorafenib maintenance	disease-free survival	not reported			not reported
37704613	sorafenib maintenance vs no sorafenib maintenance	risk of graft-versus-host disease	not reported			not reported
40577201	sorafenib maintenance vs no sorafenib maintenance	relapse-free survival probability	SORA group 95.5% vs control group 66.7% (p=0.004)			median follow-up 20.7 months
40577201	sorafenib maintenance vs no sorafenib maintenance	median overall survival	SORA group 36.3 months vs control group 11.6 months (p=0.01)			median follow-up 20.7 months
40577201	sorafenib maintenance vs no sorafenib maintenance	mortality rate	4.2% in SORA group vs 50% in control group (p<0.001)			at last follow-up
40577201	sorafenib maintenance vs no sorafenib maintenance	risk of graft-versus-host disease	not reported			not reported
32791048	sorafenib maintenance vs non-maintenance	1-year cumulative incidence of relapse	HR 0.25 (95% CI 0.11-0.57)	0.25	0.11–0.57	1-year; median follow-up 21.3 months
32791048	sorafenib maintenance vs non-maintenance	overall survival	not reported			not reported
32791048	sorafenib maintenance vs non-maintenance	acute graft-versus-host disease	23% (23/100) in sorafenib group vs 21% (21/102) in control group			Within 210 days post-transplantation

PMID	Comparison	Outcome	Effect	Value	95% CI	Follow-up
32791048	sorafenib maintenance vs non-maintenance	chronic graft-versus-host disease	18% (18/99) in sorafenib group vs 17% (17/99) in control group			not reported

Synthesis of findings

Nine studies investigated tyrosine kinase inhibitor (TKI) maintenance therapy in adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who had undergone allogeneic stem-cell transplantation. These studies evaluated gilteritinib, sorafenib, or quizartinib maintenance against comparators such as placebo, no TKI maintenance, no sorafenib maintenance, or non-maintenance control. All included studies directly addressed the PICO population, intervention, and comparator.

The evidence generally indicates a benefit of TKI maintenance therapy on survival outcomes. For relapse-free survival (RFS), sorafenib studies consistently reported hazard ratios (HRs) favoring the intervention (e.g., HR 0.39, 95% CI 0.18-0.85; HR 0.47, 95% CI 0.30-0.73). Gilteritinib showed a trend towards improved RFS (HR 0.679, 95% CI 0.459-1.005) in one study. For overall survival (OS), quizartinib and sorafenib studies reported HRs ranging from 0.50 to 0.65, indicating improved OS with TKI maintenance.

Regarding graft-versus-host disease (GVHD), most studies did not report comparative effect estimates. One study on sorafenib ^[1] reported no significant difference in 5-year cumulative incidence of chronic GVHD (HR 0.82, 95% CI 0.56-1.19), and another ^[2] showed similar rates of acute and chronic GVHD between groups.

Limitations include some studies not providing specific effect estimates for all outcomes or for the full cohort, and variations in follow-up durations across the included evidence.

Discussion

Principal findings and implications

The synthesis presented in the Results section integrates the available evidence on the research question and characterises the consistency, magnitude, and direction of the reported effects. These findings should be interpreted in the context of the methodological considerations described below.

Strengths and comparison with other studies

The pipeline applies a reproducible, transparent search strategy with a fully audit-able query string (see the Appendix). Every screening decision and extraction artefact is preserved as a CSV record, supporting independent re-analysis. Retrieval-augmented MeSH expansion grounds term selection in actual reference-paper vocabulary rather than language-model training data alone, which reduces the risk of hallucinated terms that would inflate or deflate retrieval recall.

Limitations

Several limitations of this automated review should be noted. First, retrieval was limited to PubMed/MEDLINE; studies indexed only in Embase, the Cochrane Central Register of Controlled Trials, or grey-literature sources are not represented. Second, screening and extraction relied on language-model judgements, which may misclassify edge cases. Third, the cumulative relevance threshold is a configurable cut-off that may exclude some borderline-eligible studies. Fourth, the synthesis is qualitative; no formal meta-analysis, risk-of-bias assessment, or assessment of certainty of evidence (e.g., GRADE) was performed. Fifth, automated extraction may miss subtle methodological nuances of individual studies.

Conclusions

Nine studies directly addressing the planned comparison found that tyrosine kinase inhibitor maintenance therapy generally improves relapse-free survival and overall survival compared to no TKI maintenance in adults with FLT3-ITD-mutated acute myeloid leukemia who have undergone allogeneic stem-cell transplantation, with no clear impact on graft-versus-host disease.

Abbreviations

CIs: Confidence intervals; GRADE: Grading of Recommendations Assessment, Development and Evaluation; HRs: Hazard ratios; LLM: Large language model; MeSH: Medical Subject Headings; PICO: Population, Intervention, Comparator, Outcomes; PMID: PubMed identifier; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-analyses.

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Funding

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Availability of data and materials

All extraction artefacts (PubMed query, retrieved records, screening decisions, study characteristics, outcome data, and the synthesis narrative) are preserved on the host filesystem under run identifier 6d09b924e073 and are available for verification or re-analysis.

Authors' contributions

Synthesis generated by the REASON automated pipeline; reviewed by the user of the dashboard. Final responsibility for the content lies with the human reviewer.

Ethics approval and consent to participate

Not applicable. This review is a secondary analysis of previously published reports.

Consent for publication

Not applicable.

Competing interests

The authors of this automated report declare no competing interests.

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8. Impact of genetic patterns on sorafenib efficacy in patients with FLT3-ITD acute myeloid leukemia undergoing allogeneic hematopoietic stem cell transplantation: a multi-center, cohort study. PubMed PMID: [37704613](#).
9. Post-transplant sorafenib effectively prevents relapse in FLT3-mutated acute myeloid leukemia. PubMed PMID: [40577201](#).

Appendix A. Search strategy

The exact PubMed query string executed against the eutils esearch endpoint:

```
(FLT3-ITD AML OR acute myeloid leukemia OR  
allogeneic stem cell transplantation OR FLT3  
internal tandem duplication OR allogeneic HCT OR  
allo-HSCT OR hematopoietic cell transplantation) AND  
(tyrosine kinase inhibitor OR FLT3 inhibitor OR  
maintenance therapy OR gilteritinib OR sorafenib OR  
quizartinib OR post-transplant maintenance) AND  
(relapse-free survival OR overall survival OR  
graft-versus-host disease OR RFS OR OS OR relapse  
incidence OR GVHD) NOT (review[Publication Type] OR  
meta-analysis[Publication Type] OR systematic  
review[Publication Type])
```

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